

National College of Ireland

Project Submission Sheet

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Research Practicum (MSCODP1)

Dynamic Personalization for AI Assistants: A Comparison between Fine-Tuning with LSTM and Retrieval-Augmented Generation

Your Name/Student Number	Course	Date
Jai Prabhas Malluri - 24310875	Research Practicum	27/04/2026

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Tool Name	Brief Description	Link to tool

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[Insert Sample prompt]	[Insert Sample response]

Evidence of AI Usage

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

Additional Evidence:

[Place evidence here]

Additional Evidence:

[Place evidence here]

Dynamic Personalization for AI Assistants: A Comparison between Fine-Tuning with LSTM and Retrieval-Augmented Generation

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MSc in Open Data Practice

National College of Ireland

Abstract

Our work investigates a personalised AI assistant which can interpret intent and respond with context-aware generation on a knowledge base through fine-tuned Long Short-Term Memory (LSTM) networks and Retrieval-Augmented Generation (RAG). Large language models (LLMs) are conversational but they don't generally retain user information without lengthy retraining. This research (1) implements a baseline intent-classifier using the Banking77 dataset with traditional machine learning classifiers and an LSTM model, and (2) contextualises generation by running a RAG pipeline with Google's Agent Development Kit (ADK) and Gemini models to generate information relevant to the user's queries from a dynamic knowledge base, preventing hallucinations. We evaluate these methods (we compare them on accuracy and F1 for classification, and ROUGE, BLUE and Exact match for generation) to help design scalable and private AI assistants.

Keywords: Retrieval-Augmented Generation, LSTM, Fine-Tuning, Google ADK, AI Assistants, Personalisation

1. Introduction

The rapid development in AI technologies has resulted in increasing sales of personal assistants where users expect a machine to understand their voice requests, and to provide them with information in financial, health, smart home automation and productivity scenarios (Adamopoulou and Moussiades, 2020). This has been achieved in two ways: the use of language models based on transformer architecture and information retrieval systems based on transformers (Adamopoulou and Moussiades, 2020).

There are two main approaches. Fine-tuning adapts an existing model to a user's data to store knowledge in model weights (Howard and Ruder, 2018) but this can be slow and expensive for user-specific knowledge, and make updates or removal of storage challenging due to catastrophic forgetting (Kirkpatrick et al., 2017). Retrieval-Augmented Generation (RAG) instead retrieves

knowledge on-the-fly from an external knowledge source, which can be changed or refreshed (Lewis et al., 2020), making updates faster and improving factuality by combining responses to matches with knowledge source.

This article therefore is in two parts. First, it establishes a fine-tuning baseline, fine-tuning traditional classifiers and an LSTM (long short-term memory) network on the Banking77 intent benchmark (Casanueva et al., 2020), a very small dataset that can be experimented with on low-end hardware. Then it explores RAG-as-a-response-pipeline through the Google Agent Development Kit (ADK) (Google, 2025) that combines retrieval and generation using Gemini models that access an up-to-date knowledge base. We experiment with these two approaches by comparing accuracy of generated responses, as well as knowledge base updatability.

This work aims to answer two questions: (1) Is RAG more effective in terms of response accuracy and costs than fine-tuning of a rapidly-updated assistant? (2) Can we employ a hybrid approach with LSTM fine tuning for intent classification, and RAG for variable information retrieval and generation?

This paper adds value by: (1) performing experimental comparisons between fine-tuning and RAG in terms of classification and generation; (2) proposing a hybrid approach using efficient fine tuning for classification and RAG for variable information retrieval and generation using Google ADK, which operates on low cost hardware; and (3) discussing ethical considerations of personalised AI in terms of privacy and consent, and misuse of information to propagate targeted misinformation from GDPR-relevant perspectives.

The remainder of the paper discusses our literature review (Section 2) and research plan (Section 3) including the evaluation plan for our proposed project.

2. Literature Review

Language assistants are at the intersection of information retrieval, natural language processing and generative AI. We are inspired by the approaches of (A) RAG for dynamic knowledge retrieval, (B) fine-tuning LSTM for user intent recognition and (C) conversational assistants, which we combine with "large language models" (LLM) to create the hybrid approach explained in this study.

A. Dynamic Knowledge Retrieval via Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) (Lewis et al., 2020), uses a pre-trained generator and a dense retriever (e.g., DPR) that grounds generation on evidence documents retrieved at retrieval time. They show that retrieval grounding improves performance on knowledge-intensive tasks and

improves the factual groundedness via evidence sourced from a document store, in addition to that stored implicitly in the model's weights. This has obvious applications for personal assistants that use personal facts that must be introduced without retraining models; but it was initially applied to models that know general knowledge rather than personal knowledge bases that might change over time.

Gao et al. (2024) have surveyed RAG and described three stages, Naive RAG (retrieve-then-generate, with the retrieved text Concatenated to the input prompt), Advanced RAG (improvements, such as rewriting, rerank, compression) and Modular RAG (separable, optimisable stages in the RAG pipeline). Such developments are important for assistants as facts about users are diverse and dynamic so high quality evidence retrieval and selection is required. The survey also includes evidence that mixed retrieval (combination of sparse lexical and dense semantic retrieval) outperform each technique alone, further suggesting a combination of approaches for assistants.

Edge et al. (2024) also proposed using Graph RAG which can build a graph for knowledge and support local and global retrieval using community structure. This is attractive for personalisation as people see knowledge as relational (people, events, attributes and preferences). But constructing and analysing the graph can be costly, potentially slowing system performance for conversational assistants.

Retrieval can assist dialogue systems to avoid hallucinations, rating better in perceived informativeness and coherence (Shuster et al. 2021). For conversational assistants, this gives confirmation of using a simple retrieval approach as an augmentation to ensure information about the user is always included in the response, even if not with more structured information (e.g., graphs).

B. LSTM and Fine-Tuning for Natural Language Understanding

Long Short-Term Memory networks (LSTMs) (Hochreiter and Schmidhuber, 1997) are recurrent networks that try and remember far-distant information by using input, forget and output gates and thereby solving the vanishing-gradient problem. Transformers have proven to be the model of choice for high accuracy across many NLP tasks, but the use of LSTMs is often justified by their fast training times, effectiveness with moderate data sizes and the ability to run on a CPU.

Casanueva et al. (2020) recently released Banking77, a corpus of 13,083 customer utterances in customer service with 77 intents. Its size makes it a good choice for local experiments, and its structure allows comparisons between small classifiers and an LSTM baseline, the focus of this

work when considering real-world constraints of model deployment, rather than performance.

The models used for text classification include linear (Joachims, 1998), ensemble (Breiman, 2001) and transformers (Devlin et al., 2019). Transformer-based intent models can be highly effective on Banking77, but may require more resources. To supplement the supervised classifiers, to offer a more commercially real baseline model that takes into account the low-resourced settings in which text classifiers are frequently used, we therefore use an LSTM in our study.

Howard and Ruder (2018) developed ULMFiT and showed that language models can be easily fine-tuned for classification with techniques such as discriminative fine-tuning and progressive unfreezing. Their research motivates LSTM fine-tuning as a suitable task-specific baseline approach, which we use for intent classification.

C. Architectures and Research Gap for Personalised AI Assistants

In recent research, there is a growing trend to think of assistants as agent-based systems. Packer et al. (2024) developed MemGPT, where a memory hierarchy is created that is able to handle a fixed context size with information that can be moved from long-term to short-term memory. This expands memory management for LLMs but doesn't specifically support online personal facts editing and linking responses to a source of truth.

Wang et al. (2024) developed a personal knowledge graph and implemented it with RAG to support inserting, deleting and updating personal facts. It does the job but adds complexity to the system (e.g., working with or navigating the graph). Our research asks if it's possible to provide a similar degree of personalisation with a more efficient and better engineered RAG pipeline that takes up less engineering and computing resource.

Google's Agent Development Kit (ADK) 2025 is an open source toolkit for agentic and multi-agent development with the capability to use tools, maintain memory and orchestrate workflows (Google, 2025). Careful integration with Gemini and local deployment make this a good choice to experiment with a modular RAG workflow with a lower cost of off-the-shelf hardware, as is the case here.

Overall, existing works have enhanced Retrieval Augmented Generation for generic knowledge-based tasks and fine-tuning for task-specific NLP, but less attention has been paid to practical, end-to-end comparisons. This paper helps address this by proposing and analysing a hybrid solution that uses the strengths of LSTM-based systems to recognise intent, and ADK-based ones for RAG to dynamically generate evidence-based responses.

3. Research method and specification

This chapter describes our method for addressing these research questions, the hybrid design, the design of the experimental system, evaluation methods, ethics and system implementation plan.

A. Research Method

Our study uses an experimental method to investigate the approaches by using the cross-industry standard process for data mining (CRISP-DM) (Wirth and Hipp, 2000) business understanding, data preparation, modelling, evaluation and deployment analysis. This understanding helps us to enhance the fine-tuned intent classification and the RAG method.

Our study has two phases. We first set a baseline for intent classification on the Banking77 dataset (13,083 utterances in 77 labels) using traditional (Logistic Regression and Random Forest) and deep learning (LSTM) models. We tokenise and normalise the utterances, use TF-IDF for traditional models and embeddings for the LSTM model. We report various metrics (accuracy, precision, recall and F1) and the time needed to train/compile the models. In this case, this component of our system is a baseline model.

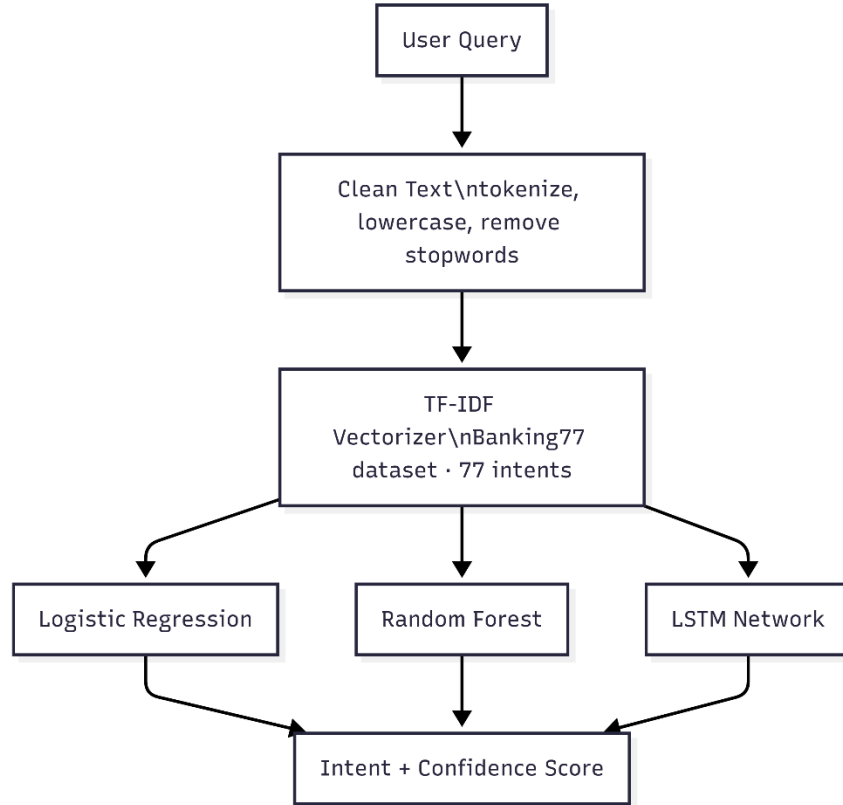


Figure 1: Proposed Fine-Tuning Architecture for Intent Classification

Second, we build a response pipeline with Google ADK to generate responses with RAG. We

encode a knowledge base (FAQ style) with Sentence-BERT and store it locally in a vector database (ChromaDB). Every time a user poses a question, we retrieve the most relevant passages and pass the retrievals to a fine-modelled Gemini model to get an evidence-based response (ADK's session memory helps keep track of context between responses). Uplift and fallback processes ensure correct, more evidence-based "report card" style answers.

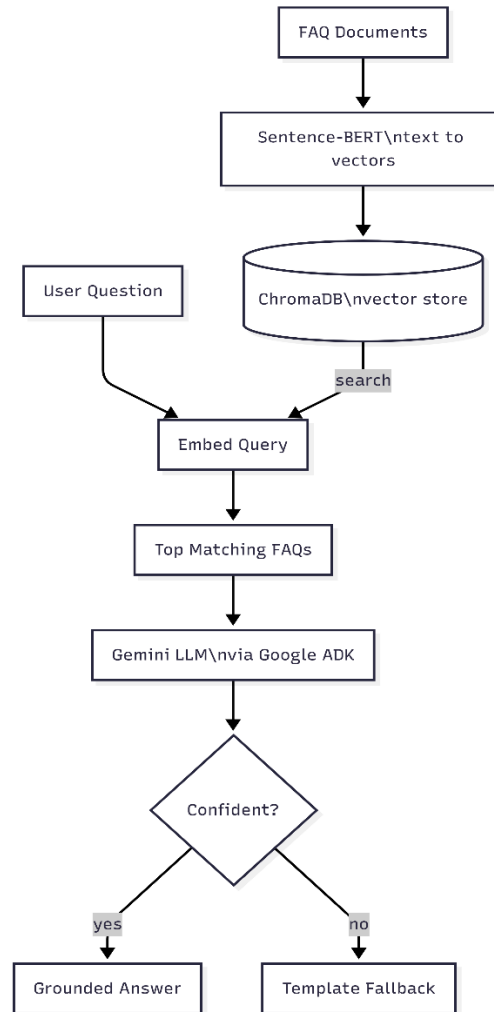


Figure 2: Proposed RAG Architecture with Google ADK & Gemini

B. Research Resources

Our concept is meant to run on off-the-shelf computing hardware. The model uses Python and scikit-learn (Logistic Regression, Random Forest) and Keras/TensorFlow (LSTM). The RAG model uses ADK to coordinate the models, Sentence-Transformers for embeddings, ChromaDB for the vector database and Gemini for generation. The Banking77 dataset is pulled from Hugging

Face Datasets, with the RAG data being a few FAQs.

C. Evaluation

Our concept is meant to run on off-the-shelf computing hardware. The model uses Python and scikit-learn (Logistic Regression, Random Forest) and Keras/TensorFlow (LSTM). The RAG model uses ADK to coordinate the models, Sentence-Transformers for embeddings, ChromaDB for the vector database and Gemini for generation. The Banking77 dataset is pulled from Hugging Face Datasets, with the RAG data being a few FAQs.

D. Ethics and Research

Ethics stems from privacy issues and security over personalised data in GDPR. Our research eliminates this concern by using a benchmark (Banking77) dataset for classifications and mocked FAQ documents for RAG's knowledge base, and thus avoids gathering personal data.

Secondly, natural hallucination. This is addressed by RAG; the model's response is based on the evidence, and the prototype additionally uses verification and confidence scores, and uses alternative answers based on low confidence and evidence-based answers.

Thirdly, is bias. We examine per-intent accuracy to understand differences in accuracy across classes that could be alleviated by class weighting and downsampling in follow-up work.

E. Project Plan

We will complete this project in 16 weeks in four phases. Weeks 1-4: complete the literature review, establish Banking77, build the FAQ Knowledge base and gain access to ADK and Gemini. Weeks 5-8: develop and evaluate the baselines for classification. This is when we will build the RAG and measures such as prompt safety. Weeks 13-16: build and compare accuracy, latency, and adaptability, write report.

Activity	W1– 4	W5– 8	W9– 12	W13– 16	Deliverable
Literature review, data preparation, and ADK environment setup	X				Prepared datasets, knowledge base, and development environment
LSTM and ML baseline training on Banking77		X			Trained classification baseline with metrics

RAG pipeline using Google ADK, Gemini, and ChromaDB			X		Functional RAG assistant prototype
Comparative evaluation, integration, and report writing				X	Final report and code submission

4. References

1. Adamopoulou, E. and Moussiades, L. (2020) 'An overview of chatbot technology', in Maglogiannis, I., Iliadis, L. and Pimenidis, E. (eds.) Artificial Intelligence Applications and Innovations. AIAI 2020. IFIP Advances in Information and Communication Technology, vol. 584. Springer, pp. 373–383.
2. Breiman, L. (2001) 'Random forests', Machine Learning, 45(1), pp. 5–32.
3. Casanueva, I., Temcinas, T., Gerz, D., Henderson, M. and Vulic, I. (2020) 'Efficient intent detection with dual sentence encoders', in Proceedings of the 2nd Workshop on Natural Language Processing for Conversational AI, pp. 38–45.
4. Devlin, J., Chang, M.-W., Lee, K. and Toutanova, K. (2019) 'BERT: Pre-training of deep bidirectional transformers for language understanding', in Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT), pp. 4171–4186.
5. Edge, D., Trinh, H., Cheng, N., Bradley, J., Chao, A., Mody, A. and Larson, J. (2024) 'From local to global: A graph RAG approach to query-focused summarization', Technical Report, Microsoft Research, arXiv:2404.16130.
6. Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Guo, Q., Wang, M. and Wang, H. (2024) 'Retrieval-augmented generation for large language models: A survey', arXiv preprint arXiv:2312.10997.
7. Google (2025) Agent Development Kit (ADK): Open-source framework for building multi-agent AI systems. Available at: <https://google.github.io/adk-docs/> (Accessed: 26 April 2026).
8. Hochreiter, S. and Schmidhuber, J. (1997) 'Long short-term memory', Neural Computation,

9(8), pp. 1735–1780.

9. Howard, J. and Ruder, S. (2018) 'Universal language model fine-tuning for text classification', in Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL), pp. 328–339.
10. Joachims, T. (1998) 'Text categorization with support vector machines: Learning with many relevant features', in Proceedings of the 10th European Conference on Machine Learning (ECML), pp. 137–142.
11. Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., Milan, K., Quan, J., Ramalho, T., Grabska-Barwinska, A., Hassabis, D., Clopath, C., Kumaran, D. and Hadsell, R. (2017) 'Overcoming catastrophic forgetting in neural networks', Proceedings of the National Academy of Sciences, 114(13), pp. 3521–3526.
12. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T. and Riedel, S. (2020) 'Retrieval-augmented generation for knowledge-intensive NLP tasks', Advances in Neural Information Processing Systems, 33, pp. 9459–9474.
13. Packer, C., Fang, V., Patil, S. G., Lin, K., Wooders, S. and Gonzalez, J. E. (2024) 'MemGPT: Towards LLMs as operating systems', in Proceedings of the 2024 International Conference on Learning Representations (ICLR).
14. Shuster, K., Poff, S., Chen, M., Kiela, D. and Weston, J. (2021) 'Retrieval augmentation reduces hallucination in conversation', in Findings of the Association for Computational Linguistics: EMNLP 2021, pp. 3784–3803.
15. Wang, Z., Li, Z., Jiang, Z., Tu, D. and Shi, W. (2024) 'Crafting personalized agents through retrieval-augmented generation on editable memory graphs', in Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP), pp. 4891–4906.
16. Wirth, R. and Hipp, J. (2000) 'CRISP-DM: Towards a standard process model for data mining', in Proceedings of the 4th International Conference on the Practical Applications of Knowledge Discovery and Data Mining, pp. 29–39.